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Assignment No: 17

Problem Statement: To accept the number of terms and find the sum of the sine series.

Objective:
- To understand the concept of mathematical series in programming
- To implement sine series using C programming
- To practice loops and arithmetic operations.
- To learn the use of factorial and power calculations in C.

Theory: In mathematics, the sine function can be expressed as an infinite series called the Taylor series expansion. The sine series is represented as:

$\sin(x) = x - x^3/3! + x^5/5! - x^7/7! + \ldots$

Platform: Linux

Input: Value of x and number of terms.

Output: Displays the sum of the sine series.

Sample Input: Enter value of x (in radians):)
Enter number of terms: 5

Sample Output: Sum of sine series = 0.841471

Conclusion: Thus, the C program successfully calculates the sum of the sine series for the given number of terms.

FAQs:

1. What is a sine series in mathematics?
A. a) A sine series is a type of fourier series where a functions is expressed as a sum of sine functions only.
b) To represent functions that are odd.

2 Why is factorial used in sine series?
A. a) This series is derived using the Taylor series, which involves taking repeated derivatives of sinx.
b) Because factorials grow quickly, only a few terms are needed to get a very good approximations of sin x.

3. Why does the sine series have alternating signs?
A. The sine series has alternating signs because of how the function behaves when we keep taking its derivatives.

4. Which header file is required for power functions in C?
A. In C, the header file required for power functions is: #include <math.h>
The power function pow() is defined in math.h. library.

5. How does increasing the number of terms affect accuracy?
A. Increasing the number of terms in a series improves accuracy. Each term adds a correction to the previous approximation.